3.4 Hydroxy Compounds

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.4 Hydroxy Compounds
Difficulty	Medium

Time allowed: 20

Score: /12

Percentage: /100

Question 1

Which isomer of $C_4H_{10}O$ forms three alkenes on dehydration?

A. Butan-1–ol

B. Butan-2–ol

C. 2–methylpropan-1–ol

D. 2–methylpropan-2–ol

[1 mark]

Question 2

Primary alcohols can be oxidised to aldehydes using either acidified potassium dichromate (VI) or acidified potassium manganate (VII). Both these oxidising agents change colour as they are reduced.

What is the colour of each oxidising agent before and after it has reacted?

	Acidified potassium dichromate (VI)		**Acidified potassium manganate (VII)**	
	Before	**After**	**Before**	**After**
A	Green	Orange	Purple	Colourless
B	Orange	Green	Colourless	Purple
C	Orange	Green	Purple	Colourless
D	Purple	Colourless	Orange	Green

[1 mark]

Question 3

Lactic acid, $CH_3CH(OH)CO_2H$, causes pain when it builds up in muscles.

Which reagent reacts with **both** of the –OH groups in lactic acid?

A. Acidified potassium dichromate (VI)

B. Ethanol

C. Sodium

D. Sodium hydroxide

[1 mark]

Question 4

Which reagent could detect the presence of alcohol in a petrol consisting mainly of a mixture of alkanes and alkenes?

A. Na

B. Br_2 (in CCl_4)

C. $KMnO_4$ (aq)

D. 2,4–dinitrophenylhydrazine

[1 mark]

Question 5

Which reagent reacts with ethanol and also reacts with ethanoic acid?

A. Acidified potassium dichromate (VI)

B. Sodium

C. Sodium carbonate

D. Sodium hydroxide

[1 mark]

Question 6

Prop-2–en-1–ol, also known as allyl alcohol, reacts to form a product that can exist as optical isomers.

Which reagent would produce optical isomers from allyl alcohol?

A. Phosphorus pentachloride

B. Sodium

C. Bromine

D. Hydrogen and nickel

[1 mark]

Question 7

In its reaction with sodium, 1 mol of a compound **X** gives 1 mol of H_2 (g).

Which compound might **X** be?

A. $CH_3CH_2CH_2CH_2OH$

B. $(CH_3)_3COH$

C. $CH_3CH_2CH_2CO_2H$

D. $CH_3CH(OH)CO_2H$

[1 mark]

Question 8

Use of the Data Booklet is relevant to this question

Which volume of oxygen measured at room temperature and pressure is needed for complete combustion of 0.1 mol of propan-1-ol?

A. 10.8 dm^3

B. 12.0 dm^3

C. 21.6 dm^3

D. 24.0 dm^3

[1 mark]

Question 9

Which statements about 2-methylbutan-1-ol are correct?

1 It can give HCl (g) on reaction with PCl_5.

2 It can be oxidised to give an aldehyde.

3 It exists in two optically active forms.

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Question 10

Use of the Data Booklet is relevant to this question

2.30 g of ethanol were mixed with aqueous acidified potassium dichromate (VI) and the desired organic product was collected by immediate distillation under gentle warming. The yield of product was 70.0%.

What mass of product was collected?

A. 1.54 g

B. 1.61 g

C. 2.10 g

D. 2.20 g

[1 mark]

Question 11

Which alcohol gives only **one** possible oxidation product when warmed with dilute acidified potassium dichromate (VI)?

A. Butan-1–ol

B. Butan-2–ol

C. 2–methylpropan-1–ol

D. 2–methylpropan-2–ol

[1 mark]

Question 12

The compound 'leaf alcohol' is partly responsible for the smell of new-mown grass.

$CH_3CH_2CH{=}CHCH_2CH_2OH$

leaf alcohol

What will be formed when 'leaf alcohol' is oxidised using an excess of hot, acidified $K_2Cr_2O_7$ (aq)?

A. $CH_3CH_2CH(OH)CH(OH)CH_2CO_2H$

B. $CH_3CH_2COCOCH_2CO_2H$

C. $CH_3CH_2CH{=}CHCH_2CO_2H$

D. $CH_3CH_2CO_2H$ and $HO_2CCH_2CO_2H$

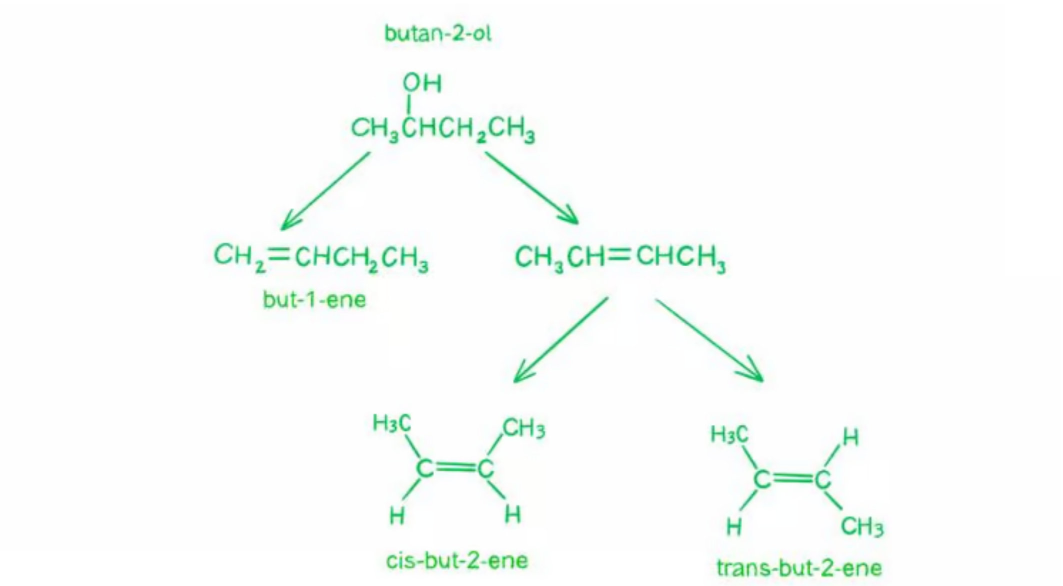
[1 mark]

Model Answers: Medium

1

The correct answer is **B** because:

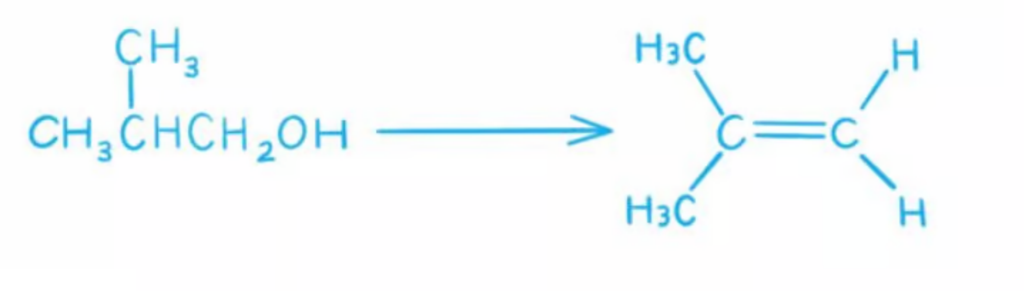
- Butan-2-ol will form 2 alkenes since it has an equal chance of producing an alkene with the double bond between the carbons 1 and 2 or carbons 2 and 3.
- Because of the placement of the double bond in but-2-ene, it exhibits cis-trans isomerism, hence it forms a mixture of **three** alkenes.



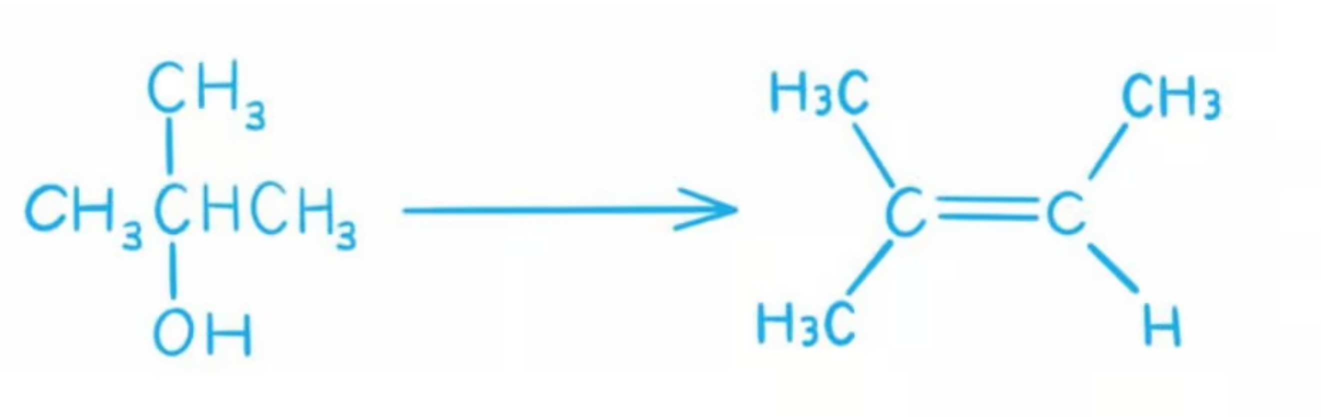
- So, the alkenes formed are:
  - But-1-ene
  - Cis-but-2-ene
  - Trans-but-2-ene

A is incorrect as butan-1-ol will form only one alkene.

C is incorrect as 2-methylpropan-1-ol will form only one alkene.



D is incorrect as 2–methylpropan–2–ol will form only one alkene.



2

The correct answer is **C** because:

- **Potassium dichromate (VI)** solution has a bright **orange** colour.
- If oxidation occurs, the orange solution containing the dichromate (VI) ions is reduced to a **green** solution containing **chromium (III)** ions.
    - So, this eliminates options **A** and **D.**
- **Potassium manganate (VII)** solution has a strong **purple** colour.
- If oxidation occurs, the purple solution containing the manganate (VII) ions is reduced to a **colourless** solution containing **manganate (II)** ions.

3

The correct answer is **C** because:

- When a **primary alcohol** reacts with **sodium**, a **salt** (sodium alkoxide) and **hydrogen gas** are formed.

  - e.g. $2CH_3OH + 2Na \rightarrow 2CH_3ONa + H_2$
- When a **carboxylic acid** reacts with **sodium**, a **carboxylate salt** and **hydrogen gas** are formed.
  - e.g. $2CH_3COOH + 2Na \rightarrow 2CH_3COONa + H_2$
- So, **1** alcohol group reacts with **1** Na atom to produce **1** hydrogen atom.
- And **1** carboxylic acid group reacts with **1** Na atom to produce **1** hydrogen atom.
- Lactic acid contains both an alcohol group and a carboxylic acid group.
  - Therefore, sodium will react with **both** of these –OH groups.

A is incorrect as acidified potassium dichromate only reacts with the alcohol to oxidise it, carboxylic acid groups are not oxidised further.

B is incorrect as ethanol will only react with the carboxylic acid group to form an ester.

D is incorrect as sodium hydroxide only reacts with the carboxylic acid in a neutralisation reaction producing a carboxylate salt and water.

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The correct answer is **A** because:

- **Alkanes** and **alkenes** do **not** react with **sodium**, whereas alcohols do.
- When an alcohol reacts with sodium, a salt (sodium alkoxide) and hydrogen gas are formed.
- If a tiny piece of sodium is added to the petrol mixture and **bubbles of hydrogen** are produced, then the liquid contains an alcohol.

B is incorrect as bromine water is used to test for the presence of a C=C double bond, so this would only confirm the presence of alkenes.

C is incorrect as potassium permanganate solution will oxidise both alcohols and alkenes.

D is incorrect as 2,4-dinitrophenylhydrazine is used as a reagent to detect the presence of aldehydes and ketones.

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The correct answer is **B** because:

- Sodium reacts with **both** alcohols and carboxylic acids.
- When ethanol reacts with sodium, sodium ethoxide and hydrogen gas are formed:
  - e.g. $2CH_3CH_2OH + 2Na \rightarrow 2CH_3CH_2ONa + H_2$
- When ethanoic acid reacts with sodium, sodium ethanoate and hydrogen gas are

- Sodium reacts with **both** alcohols and carboxylic acids.
- When ethanol reacts with sodium, sodium ethoxide and hydrogen gas are formed:
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- When ethanoic acid reacts with sodium, sodium ethanoate and hydrogen gas are formed:
  - e.g. $2CH_3COOH + 2Na \rightarrow 2CH_3COONa + H_2$

A is incorrect as acidified potassium dichromate (VI) oxidises ethanol, but not ethanoic acid.

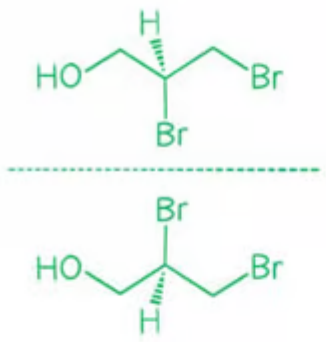
C & **D** is incorrect as sodium carbonate and sodium hydroxide react with ethanoic acid, but not ethanol – both form carboxylate salts, water and sodium carbonate also produces carbon dioxide.

6

The correct answer is **C** because:

- Prop-2-en-1-ol reacts with bromine to form **2,3-dibromopropan-1-ol.**

- Bromine will not replace the alcohol group unless it is in the form HBr where the oxygen can be protonated to create a good leaving group.
- Optical isomers are two compounds which contain the same number and type of atoms and bonds, but which have **non-superimposable mirror images.**
- 2,3-dibromopropan-1-ol forms **two optical isomers** as shown:



A is incorrect as phosphorus pentachloride is a halogenating agent. So, when combined with prop-2–en-1–ol it replaces the -OH with Cl. The resulting compound has no optical isomers.

B is incorrect as when an alcohol reacts with sodium, a salt (sodium alkoxide) and hydrogen gas are formed. The resulting compound has no optical isomers.

D is incorrect as hydrogenation of prop-2–en-1–ol leads to the breaking of the double bond. The resulting compound has no optical isomers.

7

The correct answer is **D** because:

- When a **primary alcohol** reacts with sodium, a salt (sodium alkoxide) and hydrogen gas are formed.
  - e.g. $2CH_3OH + 2Na \rightarrow 2CH_3ONa + H_2$

- When a **carboxylic acid** reacts with sodium, a carboxylate salt and hydrogen gas are formed.
    - e.g. $2CH_3COOH + 2Na \rightarrow 2CH_3COONa + H_2$
- So, 1 alcohol group reacts with 1 Na atom to produce 1 hydrogen atom.
    - The same occurs for 1 carboxylic acid group.
- If 2 form 1 mole of H_2 the compound would need to **produce 2 H atoms.**
- This can only be done if the compound has 2 alcohol groups, or 2 carboxylic acid groups, or 1 of each.
- **D** has 2 reacting groups: 1 carboxylic acid group and 1 alcohol group.
    - Therefore, option **D** is the correct answer.

A & **B** are incorrect as in both options, there is only 1 alcohol group.

C is incorrect as there is only 1 carboxylic acid group.

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The correct answer is **A** because:

- **Step 1:** Write out the equation for the complete combustion of propan-1-ol.
    - $C_3H_7OH + O_2 \rightarrow CO_2 + H_2O$
- **Step 2:** Balance the equation.
    - $C_3H_7OH + \frac{9}{2}O_2 \rightarrow \mathbf{3}CO_2 + \mathbf{4}H_2O$
- **Step 3:** Determine moles of oxygen using molar ratios.
    - For every 1 mole of propan-1-ol, 4.5 moles of oxygen are required.
    - So, 0.1 mol of propan-1-ol completely combusts in 0.45 mol of oxygen.
- **Step 4:** Multiply moles of oxygen by 24 dm^3 to determine volume occupied by oxygen.
    - Volume of oxygen = $0.45 \times 24\ dm^3 = \mathbf{10.8\ dm^3}$

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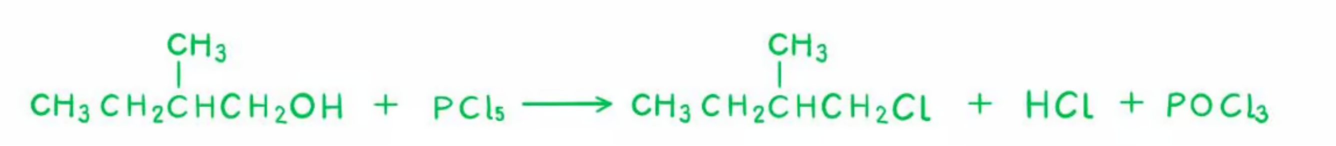
The correct answer is **A** because:

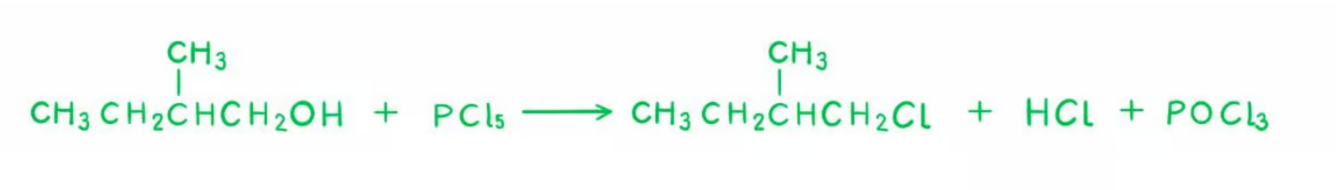
- Statement 1 is correct as 2-methylbutan-1-ol can give HCl (g) on reaction with PCl_5.



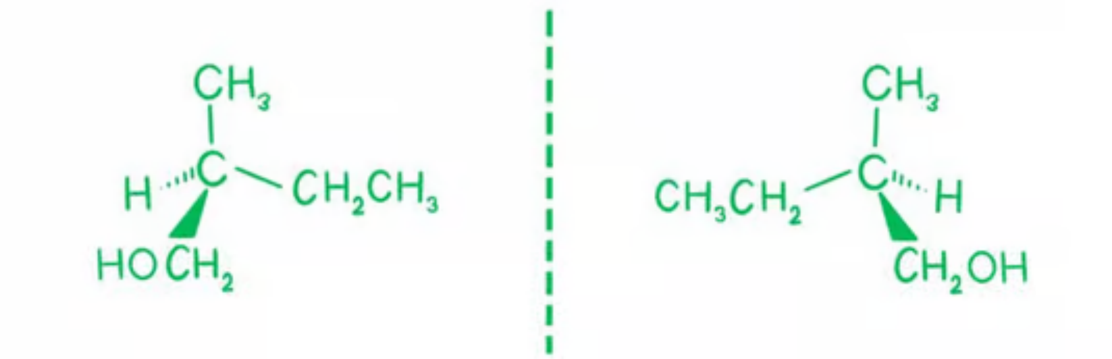
- Statement 2 is correct as 2-methylbutan-1-ol is a primary alcohol, which can be

The correct answer is **A** because:

- Statement 1 is correct as 2-methylbutan-1-ol can give HCl (g) on reaction with PCl_5.



- Statement 2 is correct as 2-methylbutan-1-ol is a primary alcohol, which can be oxidised to aldehydes or carboxylic acids.
- Statement 3 is also correct as 2-methylbutan-1-ol contains an asymmetric carbon atom (chiral carbon centre) and is therefore optically active.



- Therefore, all three statements are correct.

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The correct answer is **A** because:

- **Step 1:** Write the chemical equation
    - $C_2H_5OH + [O] \rightarrow CH_3CHO + H_2O$
- **Step 2:** Calculate the moles of ethanol
    - $Moles = \frac{Mass}{M_r}$
    - $Moles = \frac{2.3}{(12\times2 + 6 + 16)} = 0.05\ mol$
- **Step 3:** Calculate the mass of ethanol

- **Step 3:** Calculate the mass of ethanal
  - From the chemical equation, we can see that 1 mol of ethanol produces 1 mol of ethanal.
  - So, the number of moles of ethanal is 0.05 mol.

- **Step 4:** Calculate the theoretical yield of ethanal
  - Mass = Moles × M_r
  - So, the mass of ethanal = 0.05 × ((12×2) + (1×4) + 16) = 2.2 g
  - Therefore, the theoretical yield of ethanal is 2.2 g

- **Step 5:** Calculate the actual yield
  - $\% \ yeild \ = \ \frac{actual}{theoretical}$
  - So, the actual yield of ethanal = $\frac{70}{100}$ × 2.2 g = **1.54 g**

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The correct answer is **B** because:

- Primary alcohols are oxidised to aldehydes and then to carboxylic acids.
- Secondary alcohols are only oxidised to ketones, and cannot be oxidised further.
- Tertiary alcohols cannot be oxidised.
- Therefore, the only alcohol which would give only **one** oxidation product is the secondary alcohol, butan-2-ol.

A & **C** are incorrect as butan-1-ol and 2-methylpropan-1-ol are primary alcohols, so they could produce **two** products when oxidised: aldehydes and carboxylic acids.

D is incorrect as 2-methylpropan-2-ol is a tertiary alcohol, so, it cannot be oxidised.

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The correct answer is C because:

- The hydroxy group –OH in leaf alcohol is at the end of the carbon chain.
  - Therefore, it is a primary alcohol.

- When a primary alcohol is oxidised using an excess of hot, acidified potassium dichromate (VII) solution, a carboxylic acid is produced.

- The reaction looks like this:
  - $CH_3CH_2CH{=}CHCH_2\mathbf{CH_2OH} \rightarrow CH_3CH_2CH{=}CHCH_2\mathbf{CO_2H} + H_2O$
- Therefore, option **C** is correct.